Every experiment compared with the original data

How each version of the new processing step scores against the answer keys, next to what the old agent's link list contains and what the current scraper produces.

Summary

Old agent link list contains
98–100%
of answer-key families, docs, CAD and images (Bray, Crane, Sioux Chief)

Sioux Chief families found
54% → 90%
current scraper 93%

Sioux Chief families produced
4,515
real: 700; current scraper 6,324

Specs correct, any version
15–67%
no experiment moved this

- **The original data is strong.** The old agent's link list already contains nearly everything the answer keys ask for. Cimberio is the exception: its list is missing about half the main documents and the price list.
- **All the lost points are in sorting that list.** Pages get dropped by fixed rules, part numbers aren't grouped into families, and specs are misread.
- **Compared with the current scraper,** the new step is better on Bray and Cimberio grouping, Cimberio documents and images, and Sioux Chief CAD, images and sizes. It's behind on Sioux Chief main documents and image accuracy, and level on Crane overall.
- **Biggest win:** the listing-rule fix, +34 points of Sioux Chief families found. **Biggest gaps:** Sioux Chief still has about 6× too many families, and specs are 15–67% right in every version.

Every experiment, in order

"Average score" is the mean of the key accuracy measures over Bray, Crane and Cimberio. The keep rule: an experiment is kept only if that average rises and no measure on any site drops more than 2 points.

Experiment	What changed	Result	Decision
First version	A new step that sorts the old agent's link list into families, documents, images and specs.	Better grouping than the current scraper on Bray and Cimberio. On Sioux Chief, 46% of families were lost to the "listing page" rule.	Starting point
Round 1	Match part numbers as whole words; a performance chart is not a product photo.	Average score 72.1 → 74.1. Cimberio image accuracy 15% → 50%. Bray images found fell 5 points.	Kept
Round 2	Keep shared documents linked from the product's own page.	Average 74.1 → 74.4. Bray main docs 85% → 93%, other docs 81% → 95%.	Kept
Round 3	Keep documents that cover several products.	Crane other docs 55% → 94%, but specs got worse. Average 74.4 → 73.8.	Rejected, undone
Round 4	Keep an image named by the family's own series code.	Average 74.4 → 74.6. Crane images found 85% → 93%.	Kept
Crane fix	Stop reading "931" in DA931 as a size, which had merged different products.	Crane 208 → 218 families (key: 217). Scores unchanged.	Kept
Richer link list (L1)	Pretend the old agent also saved page copies and file details.	Same accuracy, 19% fewer downloads, Sioux Chief re-run 49 → 36 min. Image accuracy fell (no duplicate check without downloading).	Information only
Listing-rule fix (Option A)	Don't drop a "listing-looking" page if its web address and content say it's a product; flag it as unsure instead.	Sioux Chief families found 54% → 88%, other docs 33% → 87%. Other sites unchanged.	Kept
Item-table grouping rule	Merge two part-number pages when each lists the other in its item table.	Right 99.6% of the time. Sioux Chief 5,725 → 4,574 families, sizes found 4% → 19%, images 27% → 40%. Main docs and CAD fell because of how one split family is scored.	In the PR (formally failed the keep rule)
Claude helper	Ask Claude "are these two pages the same family?" for pairs the rules couldn't settle.	323 of 497 answered before the usage limit. Only 58% right when it said "same". 4,574 → 4,515 families.	Stopped: not the right lever

Full numbers by site

Each row is one version, measured against the same answer key. The top two rows are the original data: what the old agent's list contains, and what the current scraper produces. The last row is the latest version. Bold rows are the original data and the latest version.

Sioux Chief

% of the answer key

One web page per part number, so it is the hardest site for grouping. *The first live run used older code, and part of its higher numbers came from loose matches on series code alone. "End of rounds" is the real starting point.

Version	Families found	Families correct	Families produced (key: 700)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	100%	–	–	–	100%	100%	100%	100%	–	–	–	–
Current scraper original tool	93%	100%	6,324	0.10	46%	90%	14%	14%	79%	3%	15%	29%
First version live run, older code*	59%	100%	3,304	–	36%	17%	86%	37%	50%	9%	20%	37%
End of rounds re-run from saved copies	54%	100%	3,749	–	43%	33%	71%	22%	46%	6%	17%	34%
Richer link list (L1) simulated	54%	100%	3,749	–	43%	33%	71%	21%	33%	6%	17%	34%
Listing-rule fix (Option A) kept	88%	100%	5,725	0.11	43%	87%	71%	27%	50%	4%	18%	35%
Item-table grouping rule in PR	90%	100%	4,574	0.14	39%	87%	57%	40%	61%	19%	16%	32%
Claude helper (323 of 497 questions) stopped	90%	100%	4,515	0.14	39%	87%	57%	40%	62%	20%	16%	32%

Bray

% of the answer key

Almost nothing changed after round 2. The remaining misses are two families dropped as listing pages and two combined product photos.

Version	Families found	Families correct	Families produced (key: 103)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	100%	–	–	–	100%	100%	100%	98%	–	–	–	–
Current scraper original tool	97%	82%	122	1.00	93%	95%	100%	80%	85%	100%	67%	86%
First version live run	97%	94%	107	–	85%	81%	100%	78%	84%	100%	67%	85%
Round 1 kept	97%	94%	107	–	85%	81%	100%	73%	86%	100%	67%	85%
Round 2 kept	97%	94%	107	–	93%	95%	100%	73%	86%	100%	67%	86%
Round 3 rejected	97%	94%	107	–	93%	95%	100%	73%	86%	100%	67%	86%
Round 4 + Crane fix kept (end of rounds)	97%	94%	107	–	93%	95%	100%	73%	86%	100%	67%	86%
Richer link list (L1) simulated	97%	94%	107	–	93%	95%	100%	73%	83%	100%	67%	86%
Listing-rule fix (Option A) kept	97%	94%	107	1.00	93%	95%	100%	73%	86%	100%	67%	86%
Item-table grouping rule in PR	97%	94%	107	1.00	93%	95%	100%	73%	86%	100%	67%	86%

Crane

% of the answer key

The Crane fix stopped wrong merges (for example DA931 with DA941), taking families from 208 to 218 against 217 in the key. Round 3 raised "other docs" but made specs worse, so it was undone.

Version	Families found	Families correct	Families produced (key: 217)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	99%	–	–	–	100%	100%	67%	100%	–	–	–	–
Current scraper original tool	100%	100%	218	0.99	100%	71%	100%	78%	100%	74%	62%	83%
First version live run	100%	100%	208	–	100%	55%	100%	85%	100%	74%	62%	83%
Round 1 kept	100%	100%	208	–	100%	45%	100%	85%	100%	74%	61%	79%
Round 2 kept	100%	100%	208	–	100%	55%	100%	85%	100%	74%	62%	83%
Round 3 rejected	100%	100%	208	–	100%	94%	100%	85%	100%	74%	62%	83%
Round 4 + Crane fix kept (end of rounds)	100%	100%	218	–	100%	55%	100%	93%	100%	74%	62%	83%
Richer link list (L1) simulated	100%	100%	218	–	100%	55%	100%	93%	100%	74%	62%	83%
Listing-rule fix (Option A) kept	100%	100%	218	0.99	100%	55%	100%	93%	100%	74%	62%	83%
Item-table grouping rule in PR	100%	100%	218	0.99	100%	55%	100%	93%	100%	74%	62%	83%

Cimberio

% of the answer key

The old agent's list is missing about half the main documents and the price list, so 54% is the ceiling for main docs whatever we do.

Version	Families found	Families correct	Families produced (key: 32)	Duplicates score	Main docs	Other docs	CAD	Images found	Images correct	Sizes found	Specs correct	Specs complete
Old agent link list original data	100%	–	–	–	54%	100%	0%	100%	–	–	–	–
Current scraper original tool	97%	66%	47	1.00	25%	12%	67%	0%	–	0%	34%	43%
First version live run	100%	100%	23	–	54%	100%	100%	100%	15%	0%	53%	72%
Round 1 kept	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Round 2 kept	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Round 3 rejected	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Round 4 + Crane fix kept (end of rounds)	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Richer link list (L1) simulated	100%	100%	32	–	54%	100%	100%	100%	50%	0%	53%	72%
Listing-rule fix (Option A) kept	100%	100%	32	1.00	54%	100%	100%	100%	50%	0%	53%	72%
Item-table grouping rule in PR	100%	100%	32	1.00	54%	100%	100%	100%	50%	0%	53%	72%

How to read the measures

Families found Share of the answer key's product families that we produced.

Families correct Share of our families that are real. Not informative on Sioux Chief, where nearly every series is in the key.

Duplicates score 1.00 means one family per real family; 0.14 means about 7 pieces per real family. Added during the grouping work, so earlier versions have no value.

Docs, CAD, images, sizes, specs Measured on about 15 families per site that were checked by hand in detail. "Found" means we have it; "correct" means what we have is right.

Old agent link list Only whether an item appears anywhere in its list. It doesn't group, pick or read anything, so the other measures don't apply to it.

Sources: score files in harness/runs/scores (manufacturer-scraper-eval, PR #7), scored against the unchanged answer keys in harness/gold. Docs, CAD, image, size and spec measures use about 15 hand-checked families per site. The Claude helper row covers 323 of 497 questions (stopped at the owner's decision).